

Knight Moves Queries

Time limit: 1.00 s

Memory limit: 512 MB

There is a knight on an infinite chessboard. The rows and columns are 1-indexed.

Your task is to efficiently process queries of the form: when the knight starts at position (x, y) , what is the minimum number of moves the knight needs to do to reach the top-left corner.

Input

The first line has an integer n : the number of queries.

After this, there are n lines. Each line has two integers x and y : the knight position.

Output

For each query, print the minimum number of moves.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq x, y \leq 10^9$

Example

Input	Output
4	0
1 1	1
2 3	3
4 1	669
42 1337	